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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
Student Name	N. Siri Chandana Reddy
Roll Number	2520030142
Branch	CSE
Course Outcome	CO6 (0/1 Knapsack using Dynamic Programming)

Case Study Topic: Waste Sense – Smart Waste Collection & Recycling Optimization System Using 0/1

Work Done Summary:

The Waste Sense system uses the 0/1 Knapsack algorithm to optimize waste collection vehicle loading. Each waste bin contains a certain amount of recyclable material and has an associated value representing its recycling importance. Since collection vehicles have limited capacity, the system selects the most valuable combination of waste bins while staying within the vehicle's capacity constraints.

Implementation Details:

1. Create a list of waste bins with weights and values.
2. Define the vehicle capacity.
3. Apply Dynamic Programming (Memorization or Tabulation).
4. Calculate the maximum recyclable value obtainable.
5. Identify the optimal set of bins for collection.
6. Display the selected bins and total value.

Result: Screenshot/output:

```
Locations:
Collection Center
Bin A
Bin B
Bin C
Recycling Plant

BFS Traversal:
Collection Center -> Bin A -> Bin B -> Bin C -> Recycling Plant

DFS Traversal:
Collection Center -> Bin A -> Bin C -> Recycling Plant -> Bin B
```

The Waste Sense system successfully applies the 0/1 Knapsack algorithm using Dynamic Programming to select the most valuable combination of recyclable waste bins within vehicle capacity limits. The solution improves collection efficiency and maximizes recycling output.

GIT HUB LINK : https://github.com/nemaliscreddy78-glitch/DSA2

Student Signature	Faculty Signature